Investigating the Fraud and Misconduct in Legitimate Internet Economy On a Customer Complaint Platform

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1 Summary

Different forms of cybercrime from email spam, click fraud, to the most sophisticated underground economy have been studied extensively. However, fraud and misconduct in legitimate Internet economy, whose damage may not be as aggravated as regular cybercrime, has not received sufficient attention. In this paper, we performed the first in-depth empirical investigation of fraud and misconduct in legitimate Internet businesses by collecting 6.6 million customer complaints, which were filed over a time period of 45 months by 2.7 million customers against 117 thousand merchants on one of the largest customer complaint platforms in the world, as well as more than 13 million customer-uploaded images as photographic evidence. We characterized the complaints in terms of merchants, main issues, desired remedy, amount of money involved, duration, completion rate, and customer ratings. Most importantly, we were able to uncover various fraud and misconduct in different Internet businesses, some of which should have caught law enforcement's attention. The sum of the amount of money involved in these complaints is 5.4 billion US dollars. We also investigate the privacy inference of the images, and found that image content together with metadata embedded could disclose an unexpected amount of customers' personal information.

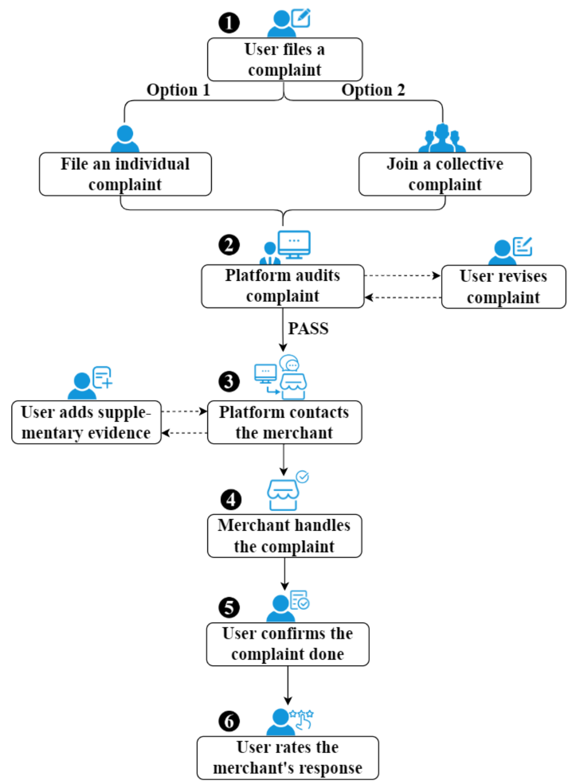

Figure 1: `BLACKCAT` overview.

2 Background

In this section, we detail the lifecycle of a complaint from being filed to completion on `BLACKCAT`, the largest online customer complaint platform in China.

The `BLACKCAT` complaint platform was launched by `Sina.com`, one Internet giant in China, in March 2018. It strives to build a third-party complaint platform for customers to directly contact merchants about a dispute. Figure 1 illustrates the lifecycle of a complaint on the platform. The ecosystem involves three parties, i.e., customers, merchants, and the platform.

When a customer files a complaint on `BLACKCAT`, he/she can either start a complaint or join a collective complaint (❶). To file a complaint, the customer needs to take two steps. (1) Customer provides relevant information including the involved merchant, issue, demand, order number and involved money. (2) the customer is required to fill in details including description of complaint and contact number, and is also encouraged to provide evidence in the form of screenshots or photos, such as order payment information, call history, and commercial contracts, to support the complaint, publicly accessible by default. Private information, containing order number and contact number, is only visible to the customer, the involved merchant, and the platform. However, it turns out that those photographic evidence is publicly accessible without any authentication. We discuss the privacy inference of those user-uploaded images in Section **??**. Collective complaint must meet the following requirements that there are at least two or more victims to complain on the same issue and against the same merchant within 72 hours.

In the next step, the platform audits the content details of complaints based on whether the business transaction is legitimate or not (❷), then publishes it on the platform. BLACKCAT also enforces a rule that any complaint published on the platform cannot be revoked or removed to prevent the case that a merchant pressures a customer to delete the complaint as a prerequisite of resolving the customer's problem. The state of the audited complaint is mainly divided into "to be allocated", "processing", "replied" and "done".

Next, the platform notifies the corresponding merchant of the complaint (❸), and then the complaint is labelled as "processing" by the platform. The platform removes a complaint if it is deemed false. Subsequently, the merchant contacts the involved customer and handles the complaint (❹), and the complaint is updated to be "replied" by the platform. Finally, a complaint is marked as "done" by the platform when either the customer confirms it done or the customer has not responded to the merchant's response within 30 days (❺). Customer also rates the response of merchant (❻).

3 Data Collection Methodology

Each complaint on BLACKCAT is assigned a unique identifier (ID), which is typically one more than the ID of the complaint preceding it. Based on the observation, we first identified two IDs, one for the first ever complaint filed on March 18, 2018, and one for the latest complaint (as of this writing) posted on December 8, 2021, and then developed an automated web crawler which accessed each complaint whose ID falls within the range delimited by the two IDs. The web crawler was developed using the Selenium WebDriver framework with python programming, and is not equipped with any functions that could bypass any CAPTCHA mechanisms or login prompts. Actually, all complaints are publicly accessible without any login required.

We conducted a comprehensive collection of the complaints, posted on BLACKCAT over a time period of 45 months (from March 2018 to December 2021), throughout the month of December 2021. Our continuous crawl went unnoticed by BLACKCAT. Specifically, our IP was never blocked during the period of measurement. Also, we were not contacted by any operator to inquire about our browsing activities. Hence we believe our crawled data is valid and not tainted by the platform operators.

Our crawled data includes complaints (individual and collective) and the profiles of merchants on the platform. An individual complaint typically consists of the following information: customer's username, merchant name, specific issue, desired remedy from merchant, amount of money involved, current status, details (date and time, text description and etc.) about each step in the lifecycle from filing to resolution (if any), and photographic evidence. A collective complaint is in essence a group of individual complaints with the same issue and against the same merchant. It contains two pieces of specific information, i.e., number of customer participants and percentage of individual complaints it contains that have been resolved (i.e., completion rate). A merchant's profile on the platform mainly contains the information about the sum of complaints received and the overall customers' rating.

# Complaints received	#Merchants	Sum of complaints	Med. complaints
>100,000	7(0.005%)	=>1,415,186 (21.3%)	192,592
10,000~100,000	106(0.1%)	=>2,515,462 (37.8%)	19,538
1,000~10,000	577(0.5%)	=>1,642,453 (24.7%)	2,069
100~1,000	2,317(2.0%)	=>719,948 (10.8%)	232
<100	114K(97.4%)	=>356,053 (5.3%)	1

Table 1: Breakdown of merchants based on the number of complaints against them.

4 Dataset Description

In total, we managed to collect a large dataset about 9,148,881 complaints, of which 6,657,560 (72.8%) complaints are publicly accessible and 2,491,321 (27.2%) are completely hidden from the public. In this study, we focus on public complaints.

Breakdown of Merchants. There are 117,243 distinct merchants involved in the public complaints in total. Table 1 provides a breakdown of merchants based on the number of complaints they received. The top 7 merchants, each of which received more than 100,000 complaints, are Taobao, JD, Pinduoduo, China Mobile, China Unicom, Meituan, and POIZON. They represent the three largest e-commerce platforms (the first three), the two largest telecommunication service providers (the middle two), and the largest food delivery platform (the second to the last) in China. In total, complaints against the 7 merchants account for 21.3% of all complaints observed. On the other hand, 71.7% of the total merchants received only one complaint, and 97.4% of merchants received less than 100 complaints during a time period of 45 months. Actually, nearly all those merchants are small and little-known companies, and the complaints against them together account for 5.3% (356,053) only of all complaints.

To investigate the characteristics of complaints in depth, we divide merchants into nine fields based on their main business, which are *E-commerce*, *Finance*, *Entertainment*, *Telecom*, *OTA (online travel agency)*, *Sharing Economy*, *Express Shipping*, *Education*, and *Food*.

Growth of Complaints. Figure 2 depicts the trend of complaints posted on BLACKCAT in each month (and in each field) from March 2018 to November 2021. Since BLACKCAT was launched in March 2018 and at first was not well known, at most about 4,000 complaints were filed a month in the first six months. Then the complaints posted on the platform grew dramatically in around November 2018, and the complaints in the

Business Field	Complaints	Desired Remedy Actions					Money (US$)		Completion Rate		Duration	User Rating
		Refund	Compensation	Apology	Explanation	Punishment	Sum	Median	Completed in any way	Completed by customers	Median (Days)	(Avg)
E-commerce	1,885,343 (28.3%)	**57.9%**	44.9%	29.2%	12.7%	27.0%	**760M**	57	48.3%	20.3%	2.3	3.64
Finance	1,707,841 (25.7%)	**36.4%**	35.0%	35.8%	30.4%	22.6%	**3.45B**	**320**	72.6%	26.5%	2.0	3.62
Entertainment	713,394 (10.7%)	**71.0%**	26.6%	14.3%	15.0%	16.9%	211M	32	67.1%	18.9%	2.5	3.43
Sharing Economy	441,950 (6.6%)	**71.5%**	19.0%	19.0%	17.6%	13.3%	306M	47.8	65.2%	26.1%	2.2	3.84
Telecom	394,296 (5.9%)	**40.6%**	26.3%	14.3%	24.2%	2.7%	33.9M	10.6	**85.7%**	**36.8%**	1.8	**4.10**
OTA	364,970 (5.5%)	**81.0%**	9.4%	12.9%	17.6%	8.3%	134M	105.3	74.4%	33.6%	1.7	3.95
Express Shipping	307,530 (4.6%)	18.6%	**49.9%**	25.9%	47.0%	10.4%	88.3M	44.5	80.5%	31.7%	2.2	3.58
Education	287,373 (4.3%)	**93.8%**	17.4%	4.9%	9.3%	5.2%	**394M**	**923.5**	65.9%	29.7%	**3.8**	3.48
Food	195,577 (2.9%)	45.3%	**50.0%**	28.0%	16.1%	23.3%	34.3M	11.5	73.5%	30.7%	1.3	3.88
Overall	6,298,274 (94.6%)*	**54.1%**	34.0%	25.3%	20.2%	19.4%	**5.4B**	51.8	68.6%	27.1%	2.1	3.68

Table 2: Statistics of complaints in each field in terms of desired remedy actions, money involved, completion rate, duration, and user ratings. *Note that it is hard to determine the main business of merchants for a small proportion of complaints, and hence we choose not to include the statistics about those complaints here.

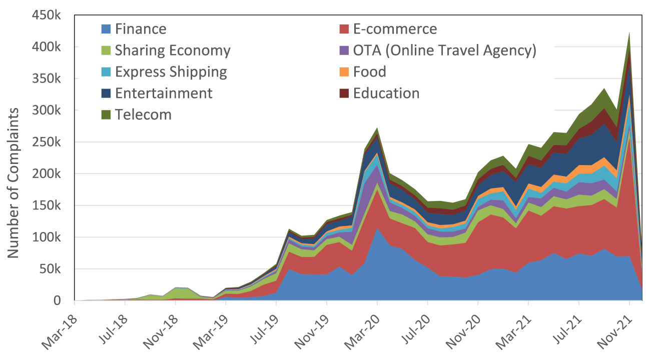

Figure 2: The variation of the number of complaints in each month from March 2018 to November 2021. Note that the data in December 2021 is incomplete since the time window of our data collection ends on December 8, 2021.

field of *sharing economy* were primarily responsible for the increase. More specifically, nearly all those complaints were against one of China's leading bike sharing start-ups, `Ofo`, which was facing serious financial problems back then [**?**].

Statistics of Complaints. Table 2 presents the statistics of complaints in each business field, in terms of desired remedy actions, money involved, completion rate, and duration from launch to completion. The 9 business fields are ordered by the complaints that fall into each field, and *e-commerce* and *finance* are the two most involved businesses.

Desired Remedy Actions. For merchants to resolve their complaints, customers usually specify remedy actions, such as refund, compensation, apology, explanation, or requesting the government punish a company severely. It appears that *refund* is the most desired remedy (54.1% of complaints) for the complaints in all business fields except in *express shipping* and *food*. For instance, up to 93.8% of complaints about education ask for a refund, indicating that nearly all such complaints are disputes over money. Most complaints in *express shipping* and *food* ask for a compensation instead.

Amount of Money Involved. A customer filing a complaint needs to provide the amount of money involved. The sum of the amount of money involved in the complaints filed on `BLACKCAT` in the past 45 months is about **5.4 billion US dollars**. *Finance*-related complaints contribute most, 63.9% of the total amount. *E-commerce* and *education*-related complaints come in second and third, accounting for 14.1% and 7.3%, respectively. As for an individual complaint, the median amount in a education-related complaint is the largest, $923.5, which will be further discussed in Section **??**, and the median amount in a finance-related complaint comes second, $320. Most complaints about *food*, *sharing economy* and *telecom* involve a much smaller amount (around $10) since these complaints are related to relatively small purchases compared to other fields. The overall median amount is $51.8.

Completion Rate. As described in Section 2, a complaint could be marked as completed either by a customer manually or by `BLACKCAT` automatically if a customer does not proceed on a merchant's response within 30 days. We distinguish the two cases since in the former case the complaint could be generally deemed resolved, but in the latter case the truth is unknown. Therefore, in Table 2, we provide the completion rates not only for the completion in any way but also for the manual completion by customers.

The overall completion rate of all complaints are 68.6% no matter how they are completed. When considering the completion performed by customers only, the rate falls to 27.1%. Complaints about *telecom* have the highest completion rates, 85.7% and 36.8%, respectively. The difference between the two is 48.9%, meaning that almost half of those complaints were closed by `BLACKCAT` due to no recent responses from customers. We argue that the completion rate may be actually somewhat higher since customers may not bother confirming the completion on the platform after their complaints are resolved.

Duration. The duration of a complaint is calculated by the difference between its completion date and filing date. Note that to be accurate we consider only the complaints whose completions were manually confirmed by the customers who filed them. The overall median duration of all complaints is 2.1 days, indicating that at least half of complaints were resolved within 2 or 3 days. Complaints about *education* have the highest median duration of 3.8 days while the median duration for the complaints about *food* is 1.3 days.

User Ratings. When closing a complaint, a customer could rate his/her satisfaction level on a scale of 1 to 5, with 1 being

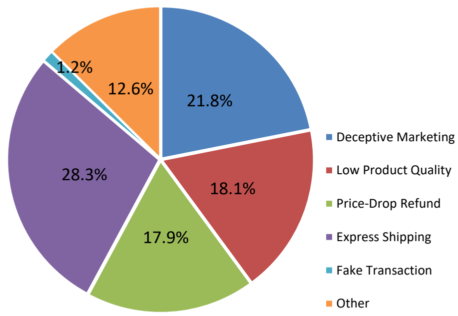

Figure 3: Top 5 main issues in e-commerce complaints.

highly unsatisfied and 5 being highly satisfied. The overall user rating score is 3.68, indicating that customers hold a somewhat positive attitude on the efficiency of the platform in resolving a dispute between customers and merchants. Specifically, customers in *telecom* complaints achieved the highest satisfaction level with the user rating score of 4.1.

Merchants' Response. In 4,923,157 (73.9%) of public complaints, merchants were found to have responded to customers' complaints against them. 1,751,487 responses were configured by merchants as hidden from others but the corresponding customers and the platform. In the rest 3,171,670 responses, 2,654 (0.8%) claimed that `BLACKCAT` made mistakes in assigning the real responsible merchant, 40,068 (12.6%) held a positive attitude and were willing to assist in solving customers' problems, and up to 2,768,348 (87.2%) responded with only empty polite words without commitment.

5 Data Analysis and Findings

In this section, we conduct an in-depth analysis of the content details of complaints, to investigate noticeable issues or scams reported in the complaints and explore the reasons behind those disputes, in order to shed light on various misconducts in legitimate economy.

5.1 E-commerce Platform

There are 1,885,343 complaints against a total of 590 e-commerce platforms in our dataset, 22 platforms of which received more than 10 thousands complaints each. The top 4 merchants receiving the most complaints are `Pinduoduo` 19.2%, `Taobao` 13.4%, `JD` 11.7%, and `POIZON` 9.0%, i.e., the three largest e-commerce platforms and the largest platform for selling fashion brands online.

Based on further analysis of the complaint contents, we identify five main issues reported in most complaints against e-commerce platforms (86.2%), which are *express shipping*, *deceptive marketing*, *low product quality*, *price-drop refund*, and *fake transaction*, sorted by the number of complaints in each category in descending order. The distribution of the five main issues is illustrated in Figure 3.

	Pindduoduo	Taobao	JD	POIZON
Express Shipping	83,528(15.6)%	96,770(18.1%)	72,110(13.4%)	57,954(10.1%)
Deceptive Marketing	130,658(31.7%)	47,809(11.6%)	41,145(10.0%)	11,030(2.6%)
Low Product Quality	62,370(18.3%)	65,252(19.2%)	47,809(14.1%)	32,465(9.5%)
Price-Drop Refund	46,034(13.6%)	44,409(13.1%)	49,166(14.5%)	26,555(7.8%)
Fake Transaction	7,350(32.7%)	5,770(25.7%)	2,435(10.8%)	583(2.6%)

Table 3: Distribution of the five main issues among the top 4 e-commerce platforms receiving the most complaints.

Express Shipping. 534,464 (28.3%) "e-commerce" complaints are about the express shipping service, as shown in Figure 3. It is reasonable since express shipping is crucial for a buyer's user experience. And 57.2% complaints about express shipping are against the top 4 e-commerce platforms, and the detailed ratios are provided in Table 3. The main issues with express shipping include lost packages, damaged packages, slow or no shipping from seller, and etc.

Deceptive Marketing. In 412,572 (21.8%) "e-commerce" complaints, customers were complaining about deceptive marketing. Table 3 shows that `Pinduoduo` received 31.7% of such complaints. Analysis of those complaint contents shows that, to attract more new customers, `Pinduoduo` rewarded loyal users with free products, cash, or other perks if the latter make bulk purchases or promote the installation of `Pinduoduo` APP via social networks. However, it turns out that many people participated in such marketing activities but they cannot receive rewards due to the crafted terms and conditions.

Low Product Quality. 340,127 (18.1%) complaints reported the issue of low product quality. Among those complaints, 25% were about clothing and 20% were about electronics. And up to 61.1% of such complaints were against the top 4 platforms. Actually, low-product-quality complaints account for more than 20% of the total complaints received by each of the four platforms. Thus, product quality is still a major challenge facing e-commerce platforms.

Price-Drop Refund. Some e-commerce platforms or sellers offer a price protection policy, which entitles a buyer to get the price difference refunded if the price is lowered on their sites within specific days (7, 15, or 30 days) after the buyer receives the item. 338,240 (17.9%) of e-commerce complaints reported that 568 platforms or sellers violate the price protection policy. The top 3 platforms accused of policy violation are `JD`, `Pinduoduo`, and `Taobao`.

Fake Transaction. E-commerce sellers may harness human laborers to conduct fake transactions and write fake reviews to enhance their store reputation and hence attract more buyers. 22,474 (1.2%) complaints were observed to accuse a total of 205 merchants of fake transactions or reviews. The content details of those complaints show that some merchants entice buyers with an extraordinary low price to place orders for fake products; some merchants ask buyers through third-party channels to participate in fake transactions voluntarily to get a cheap price; some buyers did not receive the ordered items but the express shipping status showed items were delivered already and the sellers refused a refund.

5.2 Issues across Business Fields

Till now we have studied the main issues in a specific business field, but in the rest of this section, the issues reported in complaints may exist in more than one business fields, and thus we focus on the issues rather than business fields they fall into.

5.2.1 Juvenile-related

Juveniles get special protection under the law in almost all countries. Our dataset contains 97,066 complaints involving juveniles, which were filed by at least 48,182 distinct customers against 1,493 merchants. Notably, the top 10 merchants, accounting for 47.0% juvenile-related complaints, are all online game development companies.

Further examination shows that those complaints mainly focus on the following several main issues: (1) Money deposit into game account without parents' consent. 30,682 (31.6%) complaints were filed by parents who claimed that their juvenile children deposited money into game accounts without their awareness and thus requested a refund; (2) Rewarding network anchors without parents' awareness, reported by 3,054 (3.1%) complaints; (3) Pornography and gambling information displayed to juveniles when they use smartphones for online surfing or remote learning, reported in 1,857 (1.9%) complaints.

As for the desired remedy from merchants, 87.3% complaints requested a refund, and 12.0% asked for a compensation, and 9.2% asked for an apology, and 8.0% wanted to see the merchants penalized. We can see that most of complaints involving juveniles demand to refund the money which were spent by juveniles without the consent of their parents.

5.2.2 Account Suspension

User accounts may get suspended for a certain period of time if they were deemed to have violated policies or terms of online platforms. 254,027 complaints in our dataset were about account suspension, which were filed by 129,990 distinct customers against 2,203 merchants.

The top 10 merchants receiving most of such complaints account for 41.4% complaints in total. Furthermore, those merchants fall into five categories, including e-commerce (`Taobao`, `JD` and `Pinduoduo`), online games (`TradingCat` and `Tencent Game`), video-sharing platform (`TikTok` and `Kwai`), car-sharing (`DiDi` and `DiDa`), and food-delivery (`Meituan`).

In nearly all those complaints, users believed that their accounts were suspended by mistake, and they have tried appealing account suspensions already but failed before they turn to `BLACKCAT`. And their user accounts were suspended due to various reasons. The below are our observations by looking into the complaints against several of the top merchants. (1) In about two thirds of 13,328 complaints against `TikTok`, users claimed that their `TikTok` accounts were determined to post too many videos made by others or videos containing inappropriate contents such as advertisements, and were thus suspended. (2) The 14,013 complaints against `TradingCat`, a large platform for game players to trade game accounts, props, and coins, were filed either by buyers who purchased game accounts but soon the insincere sellers or the real account owners (not the sellers) retook the control of those accounts by submitting appeals, or filed by sellers who transferred the control of accounts to buyers but then had to refund money to buyers because the latter caused the accounts to be suspended and filed an arbitration to the platform. (3) In the complaints against `Taobao`, the largest e-commerce platform in China, 1,034 complaints are filed by sellers claiming that their accounts are wrongly suspended either due to false report by buyers or for no reason, and 166 buyers claim that their accounts were suspicious of posting fake reviews and were hence forbidden to place orders or write reviews.

As for the demand of those complaints, 206,020 (81.1%) complaints demanded a revocation of account suspensions, which makes sense, and 25.8% demanded a compensation, and 23.3% wanted merchants to make apologies, and 22.5% requesteds a refund.

6 Conclusion

Previous work has studied extensively on different forms of cybercrime ranging from email spam, click fraud, to the most sophisticated underground economy. However, fraud and misconduct in legitimate Internet economy, whose damages may not be as aggravated as regular cybercrime, has been largely ignored. In this study, we performed the first in-depth empirical study of fraud and misconduct in legitimate Internet businesses by collecting a large dataset about more than 6.6 million customer complaints and more than 13 million customer-uploaded images as photographic evidence. The complaints were filed by 2.7 million distinct customers against 117 thousand distinct merchants in the past 45 months from March 2018 to December 2021. We estimated the total sum of the amount of money involved in these complaints to be 5.4 billion US dollars. We characterized the complaints in terms of the merchants, main issues, desired remedy, the amount of money, duration, completion rate, and customer ratings. Most importantly, we were able to uncover various fraud and misconduct in different forms of Internet businesses, some of which should have caught law enforcement's attention. We assess the privacy risks on those images and found that image content together with the metadata embedded could disclose an unexpected amount of customers' personal information.